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Exploring the influence of Brazilian project cancellation mechanisms on new wind power generation

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ABSTRACT

The use of wind power has gained prominence on the world stage, especially in Brazil. In addition to generating energy free of emissions, wind technology is the second cheapest generation option in Brazil. At the same time, it has contributed to reducing hydrological vulnerability caused by climate variation. Although wind power has experienced a steady expansion in the last decade, recent poor outcomes of wind auctions produced uncertainty about the future of this technology in the country. This paper analyses the influence of contract mechanisms used in Brazilian energy auctions for the growth of wind generation. To do this, a scenario analysis based on a hybrid simulation model is used, combining agent-based modelling (ABM) and system dynamics (SD). From the hybrid simulation, the effect of the wind project cancellation policy is analysed, on the one hand, and different ceiling prices for competing sources in the auctions are assessed, on the other. Moreover, the simulation model shows how hydro conditions affect reserve energy auctions.

1. Introduction

Brazil's electricity market is a large-scale hybrid renewable energy system with a predominance of large hydroelectric power plants. In recent years, the regulation in the Brazil's electricity sector has suffered several structural changes [1–4]. The aim of these changes has been to promote economic efficiency through the competitive environment, especially in the generation sector, for guaranteeing the security of supply. Since 2005, the Federal Government has designed the auctions to provide long-term agreements to build new capacity in the energy system [2,4]. Despite this, the failure to carry out projects contracted in national auctions and a decline in electricity demand prompted Brazil to create the first mechanism for cancelling reserve energy for new projects.

As several studies has proven, there are significant complementarity between hydropower and other renewable energies, including solar and wind [5,6]. Considered one of the most promising renewable technologies for energy generation, the development of wind generation technology is of particular interest for Brazil [7–10]. In addition to generating energy in a renewable and emission-free manner, the expansion of the use of this technology reduces the country's vulnerability to risks caused by hydrological variations [11,12].

Despite the growth of wind power in the last decade in Brazil, a decrease in wind auctions has provoked uncertainty about the future of this technology in the long-term [13,14]. In late 2016, the Federal Government announced the cancellation of the wind power auction that year, raising concerns in the wind industry. In the last three years, the contracts of wind energy in auctions has dropped significantly, from 4.5 GW contracted in 2017 to 3.1 GW and 1.5 GW of power contracted in 2018 and 2019, respectively [15]. Besides, projects contracted in auctions with longer lead times took more time to be completed but faced lower risk of project delays and penalties [14]. Earlier studies showed significant delays in onshore wind projects, particularly from 2005 to 2017 [16,17]. Auctions, vital for the development of the Brazilian wind industry [4,7,13,14], encourage the progress of this technology and reductions in its generation costs. Thus, the drop in implementation of this source affects the development of wind power of Brazil directly in the long-term.

To comprehend the reduction in procurement and the particularities of the deployment of wind generation technology are necessary to understand the Brazilian electricity sector. The procurement rules adopted after the division of this sector in 2004, which resulted in the creation of the Regulated Market (RM) and the Free Market (FM), have a significant influence in the expansion of the national electricity network, and

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consequently in the distribution of nonconventional renewable energy. The recent addition of a new mechanism to the existing contract modalities brings even more uncertainty regarding the expansion of the Brazilian energy matrix and the future of the generation technologies used. Created in 2017, the Reserve Energy Cancellation Mechanism is a measure taken to combat immediate problems related to the structural surplus of energy contracted in auctions. Without support in studies or previous applications in other countries, few inferences on the effects of its implementation can be found, as well as on the consequences of its use for the future of wind power in Brazil.

Given the cancellation mechanism in the electricity sector of Brazil and the importance of the development of wind power, this paper assesses the influence of guidelines adopted in the contract of electricity for wind power deployment. The guidelines analysed include the establishment of different ceiling prices in energy auctions and the adoption of the new project cancellation mechanism, in addition to particularities that govern the two Brazilian electric energy markets (i.e. RM and FM).

This paper therefore explicitly addresses the following research question: How the reserve energy cancellation mechanism affects the wind power expansion in Brazil? To do this, a scenario analysis is performed through a hybrid simulation model involving the SD and ABM approaches. The hybrid model has been proven in previous studies to outperform the SD-only model in simulating randomness and emergent issues [18]. This research chooses the SD and ABM approaches, given the recognition of using together as a solution approach to model complex energy systems [19,20]. When enough information about the individual behaviour of agents is known, ABM becomes the most suitable method to use [21]. Whereas SD is the most suitable for defining the global system behaviour [22]. Moreover, both SD and ABM are particularly suitable for generating insights into energy policy [23,24].

The paper is structured as follows. Section 2 presents an overview of the literature on the auctions and the new cancellation mechanism of Brazil. Moreover, the section provides a discussion of the hybrid modelling involving the use of SD and ABM. Section 3 discusses the methodological procedure adopted, followed by Section 4, where the simulation model is presented. Finally, in Section 5, the scenario analysis is done through simulation, followed by Section 6, where the conclusions of this paper are presented.

2. Literature review

2.1. Brazilian auctions and the new energy project cancellation mechanism

After negative experiences in the implementation of projects contracted by the alternative energy sources incentive program – PROINFA in 2002 [25–27], structural changes implemented in the country in 2004 introduced an auction system for energy contract, bringing more stability to the electricity sector. The adoption of auction-based systems has been growing, with applications in countries like the United Kingdom and China [4,28] and specific states in the United States (e.g. California) [29,30]. Traditionally, auctions reduce risks for investors because they offer guaranteed revenue for fixed periods, improve cost efficiency due to price competition, volume increase and budget control, and can increase the predictability of electricity supply from renewable energy resources [29,31,32].

The individual characteristics of auctions in each country do not allow them to be compared directly, as they are highly dependent on the context in which they are implemented [29,31]; however, international experiences allow verification of their most distinctive aspects. For instance, on the one hand, for the manager of the auction, the most significant risk is not to carry out the contracted project [29]. On the other hand, Participants face uncertainties related to the implementation costs of their projects, running the risk of submitting projects at unsustainable prices (a practice known as underbidding) [27,29].

It is important to note that even though the auctions have positive impacts, there are also negative impacts, such as the low technological diversity, low innovation rate, high administrative and transaction costs – which limits the participation of small companies – and low social acceptance [28]. These unwanted effects can be minimized by establishing auction guidelines, provided that their causes are correctly identified. Among the most discussed and implemented measures to address some of these effects are the requirement for physical and financial guarantees and the use of penalties [29]. An additional measure is the adoption of local content requirements in the implementation of energy projects [33]. Some studies also highlight the importance of the careful selection of plant construction sites, the use of regular auction schedules and the establishment of long-term contracts [14,28].

In Brazil, most of the measures found in the literature are adopted to minimize undesired results from auctions, including local content rules for granting financing [33,34], long-term contracts with penalties for non-compliance, and the requirement of physical, financial and environmental guarantees for the areas to be explored. Brazilian financial qualification levels are similar to the levels adopted in Germany and California, which establishes that investors have a rate of return on total investment of 5% [29,30]. The physical guarantee defines how much the contracted plant should contribute to the security of energy supply, is calculated by the Ministry of Mines and Energy for each project [35]. Finally, the environmental qualification is conditioned to the presentation of the prior license of the responsible environmental agency and the technical qualification by a company linked to the government (Company for Energy Research – EPE) – after analysis of the project and planned location for the construction of the wind farms – which allows a more careful selection of construction sites [4,14].

In late 2016, the non-execution of projects contracted in national auctions together with a drop in demand for electricity, which resulted in a structural surplus of power generation capacity, led to the creation of the first mechanism for the project cancellation of reserve energy, particularly for new projects. This mechanism established that projects that had not yet entered testing or operation could be cancelled without paying a fine for breach of contract, in exchange for the payment of a premium by those responsible for the projects not executed. The guidelines applied in the auction mechanism limited the participation to wind, solar photovoltaic and hydroelectric generation projects [36]. In order to penalize the uncontracted projects, it was established that the winners of the auction could not participate in two subsequent auctions for contract reserve energy, and this prohibition could be extended to the subsidiaries and controlled companies of the winning company.

Although there is no other application of a mechanism like the Brazilian auction cancellation process in other countries, the international experiences with auctions verified in the literature allow us to point out potential advantages and disadvantages of the implementation of this new mechanism. The reasons for the non-execution of contracted projects in Brazil vary, ranging from causes related to the unavailability of electric grid connections [29,37,38] to the lack of competitiveness of projects to meet established prices due to the sale at lower prices [14,28]. Many times unfinished projects in Brazil results from the overestimation of the capacity factors of the plants and the underestimation of the implementation costs, as happens in the case of France, Portugal, the United Kingdom, China and Nova Scotia [28]. The main aggravating factor for errors of this type is the use of technologies not yet fully developed or intermittent generation, such as those related to renewable sources, which justifies the focus of the Brazilian project cancellation mechanism that is applied precisely to cancel out wind projects.

Besides solving the immediate problem of the structural energy surplus, this project cancellation mechanism minimizes the risks for investors, who can resort to it in case of verification of the unfeasibility of their projects. By minimizing the investor's risks of suffering the penalties established in the procurement auctions, the mechanism also indirectly favours the undesirable practice of underbidding. The solutions pointed out in the literature to minimize the adverse effects of this

practice include the imposition of fines and the prohibition of investor participation for a certain period, both measures already adopted in the contract cancellation mechanism: the first through the payment of premiums and the second through the exclusion of investors from participation in two consecutive Reserve Energy Auctions (REAs) [13,30,35,39,40]. At the same time, minimizing investors' risks tends to increase competition at auctions, leading to falls in contract prices [14,35].

Considering that the mechanism might cancel out all the current delayed projects, the mechanism might be suspended in the future. One possibility to avoid the suspension caused by the lack of delayed projects is to broaden the scope of this auction to deconstruct projects already in operation, potentiating the effects of this mechanism. As a result, eventual structural leftovers of RM energy could be balanced more quickly, decreasing projection risks for the distributors that buy energy [13,14]. Another positive aspect of broadening the scope of this auction is to favour the expansion of renewable technologies in the free trade agreements (FTA), which retains the uncontracted projects. Besides, investors could use the funding conditions and fixed revenues made available in the regional trade agreements (RTA) to gain experience and establish themselves by migrating to the FTA. The expansion of the FM tends to increase competition in the participation of auctions for the sale of existing energy in the RM, reducing the contract prices in this market.

Migration to the FTA could boost project innovation by incentivizing more efficient power generation, considering that the values sold in the RM contracts do not have the flexibility to absorb possible increases in the generation of the ventures. In this way, the Brazilian contract mechanism has the potential to improve the efficiency of public auctions, besides serving as a reference for other countries facing similar problems of non-execution of projects contracted in auctions, such as the United Kingdom and Nova Scotia [28,41].

2.2. Hybrid models of SD-ABM

ABM and SD are simulation methods useful to understand and assess policies into complex systems [22,42–44]. Both seek to discover leverage points in complex aggregated systems [44,45], based on the assumption that a system is more than the sum of its parts and therefore needs to be viewed and studied holistically [45,46].

Both approaches are based on different premises for analysing a system. First, ABM uses a bottom-up approach, where the system is modelled from the individual entities that comprise it, while SD uses a top-down approach, analysing the system through the decomposition of its main structural components [44,45,47]. As a consequence, both methods present fundamental differences, which mark their main strengths and weaknesses [45,48]. Second, SD does not consider the behaviour at the level of agents, focusing only on dynamics at the systemic level, while ABM does not offer a systematic view and causal relationships, focusing only on dynamics at the individual level [20,44,48]. On the one hand, SD excels in the modelling of continuous aggregated systems, by reducing the diversity of the system to average aggregated values, assuming homogeneity and perfect combination within stocks and flows [49]. On the other hand, ABM stands out by modelling discontinuous properties of the system, including more considerable heterogeneity among agents, at the cost of demanding higher computational capacity [20].

The main strength of SD is its potential to infer the emergence of a particular behaviour induced by the transparency of the structure of the system, which also results in weakness as the structure of the model becomes fixed [44]. In contrast, the strength of ABM is because it can model endogenous interactions of individual agents based on decision rules, which results in weakness to the extent that it is not adequate to model factors of the macro system, such as policies [50]. It is also worth mentioning that while ABM is more effective in representing spatial and structural realism, SD models are analytically more attractive, in addition to being computationally more efficient [46].

Given the advantage and disadvantages of both approaches, a combination of SD-ABM enables to explore distinct advantages and the subjugation of the disadvantage of each methodology in the model [48]. Also, the use of a hybrid model makes it possible to overcome the computational and conceptual limitations of each approach [45], as well as to create more realistic simulation models [20].

In general, the use of hybrid models of this type provides superior results concerning their individual use, also increasing the quality of the analysis [20,45]. The SD-ABM hybridisation makes it possible to incorporate in the model, on the one hand, details and individual behaviours with simple, specific functions and rules, through the ABM. On the other hand, the SD approach enables systematic causal relationships to study and monitor the system as a whole [20,51]. At the same time, it makes it possible to increase the efficiency and transparency of modelling [45], improve computing time, capture relevant elements of individual heterogeneity and analyse the stochasticity of entities [52].

Although the use of hybrid SD-ABM models applied to different sectors have increased in the literature [20,53], few papers use this hybrid approach to assess planning alternatives for the energy sector. On the one hand, a framework of a decision support system for energy planning at national and regional levels, based on a hybrid SD-ABM model is proposed by Ref. [54]. On the other hand, a hybrid model to analyse strategies of electric energy charging of a system involving electric vehicles and micro-grids based on renewable energies is presented by Ref. [55]. Finally, a hybrid SD-ABM model for evaluating the optimal capacity of an integrated photovoltaic system and its storage units is developed by Ref. [56]. The developed model is also used to define an optimal operational decision policy considering current and future electricity market prices.

3. Method

This paper presents a hybrid model that integrates SD and ABM approaches in the AnyLogic® software. This model seeks to represent both the existing contract markets in Brazil (RM and FM) and the decisions of the different agents that comprise these markets. Thus, the hybrid model assesses the effects that may have the decisions of agents on the deployment of wind generation technology in Brazil.

3.1. Data sources

The hybrid model uses data provided by several Brazilian institutions. For instance, the information on hydrological variations in the volumes of the hydroelectric plants' reservoirs were taken from National Electric System Operator (ONS); the data on the generation of the various sources used in Brazil and connected to the National Interconnected System (NIS) was obtained from the National Electric Energy Agency (ANEEL); the wind capacity installed in the FM and RM were obtained from the Energy Trading Chamber (CCEE), and Instituto Acende Brasil provided the reports on the energy auctions held in the country.

3.2. Description of the simulation scenarios

For the creation of scenarios, two different types of policies were taken: variation of ceiling prices and replication of project cancellation policies. Both are simulated from 2018 and extend until 2030 to understand the effects of the different auction policies in the long term. Table 1 shows a summary of the policy alternatives proposed to assess the simulation scenarios.

First, the ceiling price variation policy seeks to analyse the effects of setting different price limits for competing sources in auctions. These prices, established before the execution of the auctions, determine the maximum amount to be paid by each competing source in the auction, in R$/MWh. For example, in the 3rd REA, the auction guidelines established that only projects based on hydro generation sources (Small

Table 1
Policy alternatives evaluated for the ABM-SD model.

Policy	Objective	Policy alternatives	Description
Ceiling price variation	Analyse the effects of setting different price limits for competing sources in auctions.	Free Competition	Simulate equal ceiling prices for all sources in auctions
		Incentive to other sources	Simulate lower ceiling prices for the wind source compared to other sources.
Project cancellation	Explores the effects of an expansion of the current guidelines of project cancellation auctions.	a 2 % rate of cancellation	This cancellation represented about 2% of all the capacity already contracted for the wind source in REAs
		a 15 % rate of cancellation	This cancellation represented about 15% of all the capacity already contracted for the wind source in REAs

Hydroelectric Plants - SHPs), wind and biomass could bid. The established ceiling prices were 155 R$/MWh, 167 R$/MWh and 156 R $/MWh for hydro, wind, and biomass sources, respectively. Thus, projects based on wind sources had an advantage of 7.2% over hydro and 6.6% over biomass. As a result, a SHPs could be contracted in this auction to sell the amount of energy of 1 MWh for a maximum of R$ 155.00, while a wind power plant could sell the same amount of energy for up to R$ 167.

Another example is the 3rd Alternative Sources Auction (ASA). In this auction, only plants based on biomass thermal generation or wind sources could bid, with ceiling prices of R$ 215.00/MWh and R$ 179.00/MWh for biomass and wind, respectively. In this way, the wind source competed at a disadvantage, with a ceiling price 16.7% lower than its competitor.

Considering the behaviour of the ceiling price variation policy, the simulation model considers assessing two variations: i) "Free Competition": to simulate the establishment of equal ceiling prices for all sources participating in auctions; and ii) "Incentive to other sources": to simulate the establishment of lower contract ceiling prices for the wind source compared to other competing technologies.

Second, the policy of project cancellation was drawn up based on the result of the only auction of this type that took place in the country. The 1st Reserve Energy Project Cancellation Mechanism, which occurred in 2017, cancelled a total of 557.36 MW of contracted generation capacity, of which 307.7 MW belonged to wind power plants. This cancellation represented about 2% of all the capacity already contracted for the wind source in REAs. Thus, as established in the guidelines for this type of auction, this policy is applied only to ventures contracted in the previous REAs and therefore does not apply to other types of existing auctions. Only one change is adopted, about the guidelines established in this 1st project cancellation auction: the possibility of in-operation projects cancellation. This modification is made to explore the effects of a possible expansion of the current guidelines of project cancellation auctions, affecting only projects that have not yet started their operations. This policy is simulated with two variations: i) a 2 % and ii) a 15 % rate of cancellation.

In this context, eight different scenarios were created according to the variations of the two policies above, as presented in Table 2.

The policy alternatives were assessed on a baseline scenario, which represents what would happen if the actual historical trend were to remain unchanged. By varying the parameters of the base scenario according to the policies applied, we have a new scenario corresponding to the policies applied. The simulated results are compared with the base scenario results to identify the effects of this parameter variation.

Table 2
Description of scenarios generated by policy implementation.

Description of the policies simulated	Scenario ID
Free competition	1
Incentive to other sources	2
Cancellation of 2%	3
15% Cancellation	4
Free competition and 2% disengagement	5
Free competition and 15% disengagement	6
Incentive to other sources and 2% Cancellation	7
Incentive to other sources and 15% Cancellation	8

4. Simulation model

4.1. Model structure

Two types of agents are used: regulatory and entrepreneur. Both acts within a continuous and responsive environment to their decisions, known as the contracting market, which also influences the individual decision criteria of the agents. The decision alternatives and key parameters for the agents and environment representing Brazil's electricity market are described in Table 3. The following subsections provide more details about this environment and the agents considered in the simulation model.

4.1.1. Contract markets

For both RM and FM, it is assumed that there is a potential market for each type of source available which, depending on the agents' perception of its attractiveness, may evolve positively or negatively. The evolution of this potential market, for both contract environments, is modelled on the Bass Dissemination Model [57] and extensions proposed by Ref. [22] for the deployment of technology. The Bass model considers a logistical growth for a new technology, initiated by adopters with innovative behaviour and later spread by adopters who tend to imitate the former. The adoption rates of innovators (In) and imitators (Im) are calculated by equations (1) and (2), respectively.

$$\mathrm{In} = \mathrm{E} * \mathrm{M} \tag{1}$$

$$\mathrm{Im} = t_1 * \frac{Cf}{Cl} \tag{2}$$

Where "E" represents the initial effectiveness of incentive policies, and "M" represents the totality of entrepreneurs who are willing to build a generation plant based on a particular source (i.e. the potential market for this source). The initial effectiveness of incentive policies reflects the influence of actions to promote a particular source of generation (such as ASAs, which encouraged the initial development of some renewable

Table 3
Conceptual illustration of the ABM-SD model.

Agent and environment	Behavioural options	Key Parameters
Entrepreneur	Chooses between two types of generation: wind or other sources	• Energy auctions • Construction time • Operation time of the power plants.
Regulatory	Considers three types of new energy auctions: NEAs A-3, NEAs A-5 and REAs.	• New energy auctions • Reserve energy auctions
Contract Markets	Potential market: RM and FM	• The adoption rates of innovators and imitators. • Number of entrepreneurs. • The electricity price. • Generation for source.

sources of generation, including wind power). To equation (2), "t_1" represents the coefficient of influence of entrepreneurs adopting the technology over entrepreneurs that have not yet adopted it, "Cf" represents the installed capacity of the source of generation and "Cl" represents the auctioned capacity. The potential market for the source of generation is calculated by equation (3).

$$M = Cl * EP \tag{3}$$

The variable "EP" represents the price effect on the perceived attractiveness of the technology. Thus, as the price of a given technology becomes lower and more competitive, the more entrepreneurs are willing to adopt it. This variable depends on the costs of generating the source over time and is calculated by equation (4).

$$EP = P_i * A \tag{4}$$

The variables "Pi" and "A" represent the initial price of technology generation and the learning effect on the price, respectively. The learning effect on price is the result of the learning curve gains considered, which estimate the rate of reduction in production costs that can be attributed to efficiency gains from accumulated production experiences [58,59]. This rate increases as the generation cost decreases, reaching the maximum value of 1 when the generation cost of the source is zero. The rate is fed by the adoption flow of the generation source (Ea), calculated by equation (5).

$$Ea = \left(\frac{E_t}{E_i}\right)^{-c} \tag{5}$$

The variable "C" is a negative exponent and determines how sharp the learning curve is. The variables "E_t" and "E_i" represent the accumulated experience and the initial experience. Thus, the increase of the experience on the wind source, raised to the negative exponent "C", produces a decreasing exponential curve, which also reduces the current price of the wind power exponentially.

Considering that the installed capacity of Brazil is predominantly based on hydropower, it considered that the variations inherent to this type of generation affect the deployment of other technologies, including wind technology. The monthly water generation (Gh) is calculated by equation (6).

$$Gh = ch * ec * em \tag{6}$$

Where "ch", "ec" and "em" represent the installed capacity of hydropower, the effect of hydrological conditions and the average efficiency of generation plants, respectively. The effect of hydrological conditions represents the impacts of useful volume on water generation. This useful volume, in turn, depends on annual variations in the rainfall of Brazil (va) and the monthly seasonality of the water generation (sh), as presented in equation (7).

$$ec = va * sh \tag{7}$$

It is considered that the annual variation of rainfall occurs in a stochastic way throughout the years, reaching its minimum values in the historical droughts in 2001 and 2002 and between 2014 and 2015. The water seasonality varies monthly and was defined from the historical data in the last ten years. Moreover, the wind generation (Ge) is calculated by equation (8).

$$Ge = Ca * se * Ef \tag{8}$$

Where "Ca" represents the installed capacity of wind power in RM, "se" represents the monthly wind generation seasonality and "Ef" represents the average efficiency of wind power plants. In this case, the monthly wind generation curve is defined in the same way as the water curve, using data from the last ten years.

4.1.2. Entrepreneur agent

Considering that the model analyses the deployment of wind generation, for reasons of simplification, the entrepreneur agent chooses between two types of generation options: wind or other sources. Another simplification is the establishment of the same pattern of behaviour for all entrepreneur agents. In this sense, the model does not consider different behaviour patterns that may exist among entrepreneurs, such as individual preferences for a particular source of generation, or preferences and limitations related to the location of the entrepreneur's activity.

The entrepreneurial agents act in the same way in RM and FM, with the only difference that in RM, the agents only decide to build a plant when the demand for the energy auctions exists. In both markets, only a portion of the entrepreneurial agents' population will consider adopting the wind source for the capacity construction, thus constituting the already mentioned potential market for this source. In the case of RM, the potential market for agents willing to adopt wind technology may be reduced, depending on the guidelines established in auctions.

Once the source of generation is selected, the agent builds its plant in time "t", becoming part of the SIN and producing energy in its respective contract market. In the case of RM, "t" is determined by the time set in the New Energy Auctions (NEAs) mode (e.g. NEAs A-3 for three years and NEAs A-5 for five years), while in the FTA is set for the construction of all wind power plants in 2.33 years. A portion of RM entrepreneurs may still terminate their plants before the time set by the auction policy, acting in the FM until the "t" period is completed and they enter the RM. This situation is common in the Brazilian electricity sector and confirmed in the analysis of historical data on the output of FM generation projects for entry into RM.

Wind farms operating exclusively in the FM may still migrate to the RM, provided they can win an auction. Still considering the guidelines applied in the RM energy auctions for wind power, the contracts of the wind projects are undone after 20 years of operation of the plants and this capacity is moved to the FM.

4.1.3. Regulatory agent

The regulatory agent is composed of a single agent that makes the decisions regarding energy auctions based on the current situation of the electricity sector. This agent considers three types of new energy auctions to be added to the electric matrix of Brazil: NEAs A-3, NEAs A-5 and REAs. Moreover, the historical ASAs were included within the volume auctioned in the REAs. Considering the historical average of the REAs already in operation, a lead time of 2.75 years for the contracted plants to enter service was selected.

The migration of entrepreneurs from FTA to RM takes place through Existing Energy Auctions (EEA), limited here to a rate that defines a relative value of wind projects built making this transition. Note that the Energy Project Cancellation Auctions are applied by the original auction guideline, discounting projects contracted in previous REAs exclusively.

The regulating agency calculates the capacity to be auctioned in NEAs (i.e. A-3 and A-5) and REAs from two assumptions. First, the volume of energy contracted in the NEAs – an auction process whose objective is to contract energy to meet the future increase of the demand for electric energy in Brazil – is dependent on the increase of the Brazilian GDP. This assumption considers that an increase in electricity demand is related to the economic development of the country. Indeed, significant contraction of energy demand has been observed in recent years when there was no GDP growth. Thus, the equation used by the regulatory agency to calculate the total amount of NEA is given by equation (9).

$$V1 = Ct * mp * ep * c1 \tag{9}$$

Where "Ct" is the total installed capacity of the RM, "mp" represents the average of positive changes in GDP covering the first year of the auction and the two years preceding, "ep" represents the effect of GDP and "c1" represents the new energy conversion coefficient. The value zero is considered for negative GDP values. It is also considered that the effect

of GDP frequently varies, starting from zero and having a maximum effect for GDP values close to or above 7.5%.

Second, the assumption made for REA is that this type of auction occurs to compensate for unfavourable hydrological conditions, which affects hydropower. It is worth mentioning here the lack of any explanation on the part of the government regarding the criteria adopted for contract reserve energy, highlighted in Ref. [36], which naturally makes it difficult to create projections for this type of auction. The equation used to calculate the capacity auctioned in this mode is given by equation (10).

$$V2 = Ct*v*ev*c2 \tag{10}$$

Where "v", "ev", and "c2" represent the useful volume of water in the dams, the effect of changes in useful volume and the coefficient of conversion of reserve energy, respectively; the effect of volume variations is associated to the water volume available in the dams, reaching a maximum value of 1 when this volume is equal to 0.

Once the capacity to be auctioned is determined, the regulatory agency bids and contracts several projects proportional to the projected capacity. In this sense, two assumptions have been taken for the regulator: i) the A-3, A-5 and REAs take place once a year each and ii) the wind farms built have 30 MW of installed capacity. This assumption was chosen based on Brazilian Normative Resolution 77, of 2004, which established a 50% reduction in transmission and distribution tariffs for projects based on alternative sources of up to 30 MW of capacity, which strongly influenced the wind farms installed.

4.2. Model assessment and validation

The time horizon of the model assessment is 2007–2017; the initial modelled values of the variables for 2007 are compared with the historical data. Fig. 1 presents the evolution of installed wind capacity for RM and FM, together with the historical data. This figure showed a fall of the wind power generation for RM in 2013. This fall is related to the lack of transmission lines to supply the wind power [37,38,60–62], which prevented the entry into operation of a large number of wind farms between 2012 and 2013. These projects finally became operational in 2014 [63], marking the abrupt increase seen this year. Such atypical behaviour shows how much random events can impact the correlation between simulated and historical data, especially when analysing a complex system. The fluctuations verified in the FTA are related to the already mentioned characteristic of temporary entry of entrepreneurs in this market, while their respective contracts of operation in the RM are not yet in force.

The values of hydro and wind generation of the RM simulated for the base scenario and the historical data are presented in Figs. 2 and 3. The behaviour shows the last month of data made available by ONS until the date of completion of this research (i.e. 126 months between 2007 and June 2017).

The simulation of hydro generation presents smoother curves and at a lower level of intensity, in comparison with the historical data, as can be observed in Fig. 2. It is assumed here that such behaviour is associated with the sharper differences between rainfall and useful volume of the dams assumed by the model. Fig. 3 shows the comparison of the simulated data with historical wind generation values for the same period.

The model assessment shows that hydro and wind generation have different behaviours: hydro generation is strongly influenced by stochastic factors (e.g., climate variations), while wind generation has minor variability. The different methodologies used to model the generation of these sources in Brazil, visualized in the NEWAVE model, based on a Dual Stochastic Dynamic Programming methodology used to determinate the national hydrothermal dispatch, and calculate the expectation of wind generation.

The results of the auctioned capacity compared with the historical values in the NEAs and REAs, calculated by this agent, are shown in Fig. 4. Note that although it started in 2007, the model needs one year of simulation to generate the first auctions, in 2008, the year in which historically the first Brazilian REA occurred.

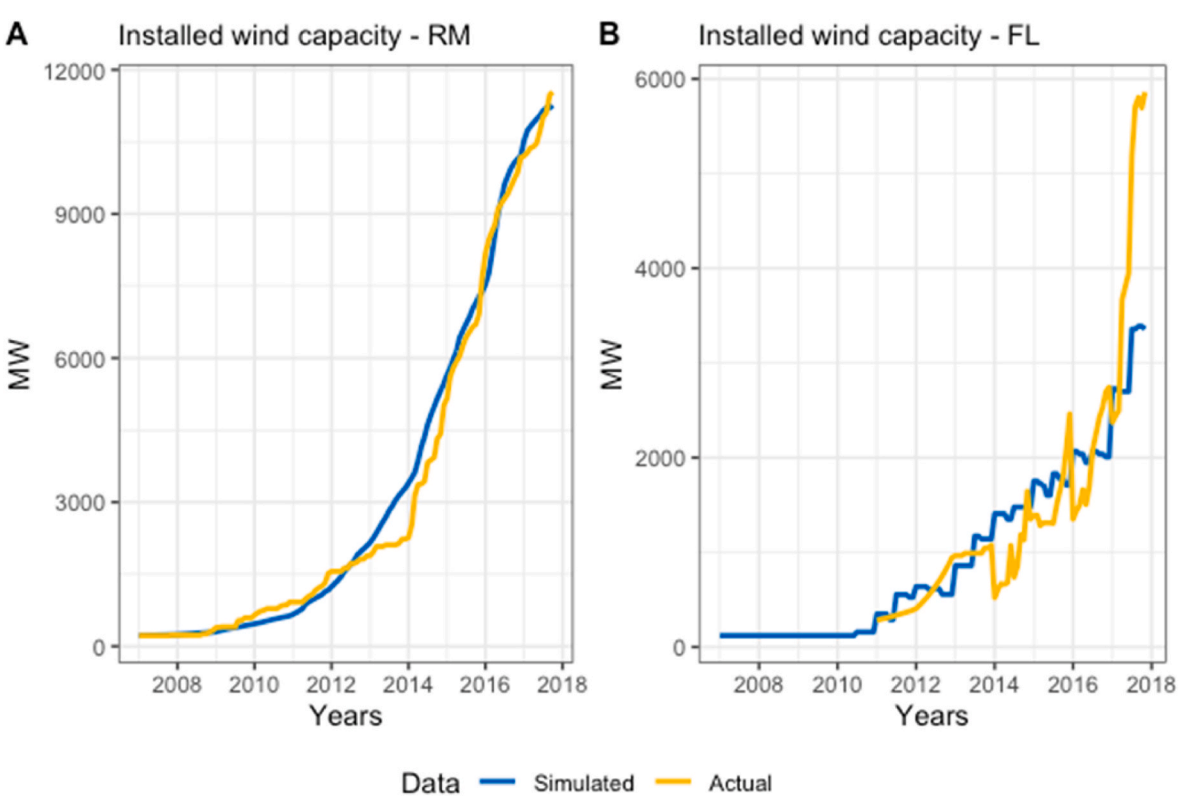

Fig. 1. Evolution of the installed wind capacity in Brazil (2007–2017).

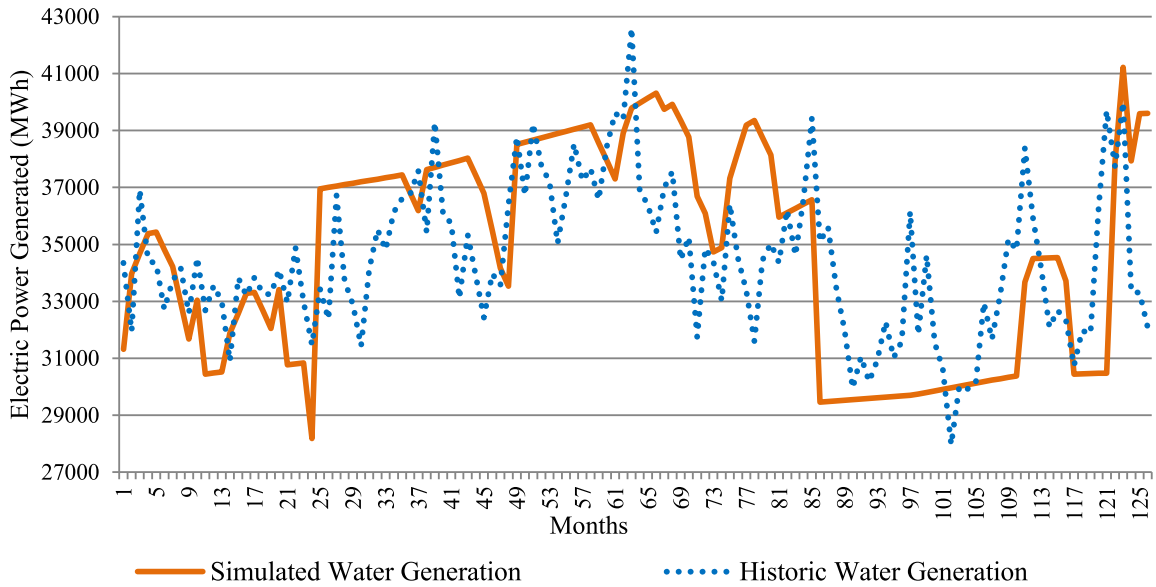

Fig. 2. Evolution of hydro generation in Brazil between 2007 and 2017.

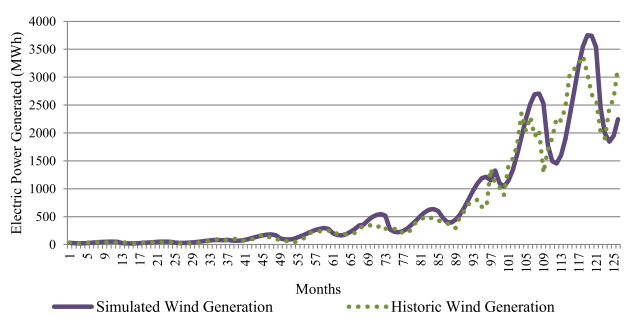

Fig. 3. Evolution of wind power generation in Brazil between 2007 and 2017.

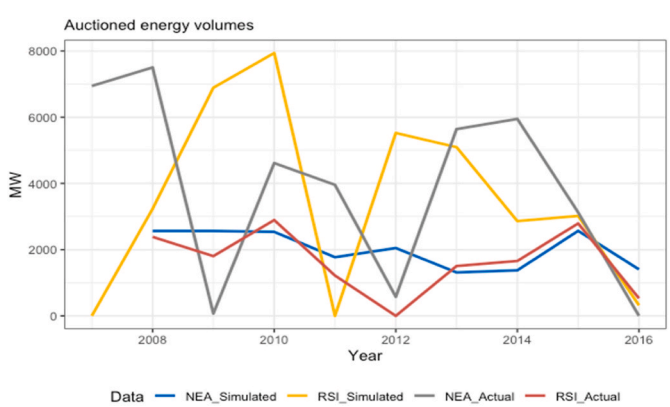

Fig. 4. Auctioned energy volumes.

There is a remarkable similarity between historical data and estimated data for REA volumes, with a significant difference only perceived in the year 2012. This year there was no REA, to which the model points out the occurrence of the capacity of reserve energy auctioned. The estimated data of capacity auctioned in NEAs follow to the historical data, but with a delay of 2 years. This behaviour is related to the auctioned capacity in the NEAs based on an average of changes in GDP in the year of the auction and in the two previous years (equation (9) presented above). A delay of two years indicates that this calculation relates better to an average involving the value of GDP in the year of the auction and estimates for the following two years (instead of historical data for the two previous years).

5. Results and discussion

5.1. Policy implementations

As mentioned above, two types of policies were selected for the creation of scenarios. The variation of the ceiling prices is applied by the regulatory agency in the establishment of the auction guidelines and directly influences the potential market. Under the "Free Competition" option, the regulating agency applies the same price ceilings to all sources competing in auctions, not influencing the reduction of the potential wind technology market. With the "Incentive to other sources" option, the regulator determines that the ceiling price of the wind source is lower than that of other competing sources. To define the values applied, simulations were made in the model until the potential market of the number of participants registered with projects involving wind power coincided with the historical data of wind projects that competed in the 22nd and 23rd NEAs, used as a reference. In these two auctions, the wind source competed with ceiling prices between 15% and 23% lower than the ceiling price set for the other competing sources.

In the policy of replication of cancellation, the percentage of wind capacity is applied to determine contracted energy, based on the total value of wind capacity contracted in REAs until the time of the auction. The uncontracted value is subtracted from this history of contracted wind capacity in REAs. It is assumed that this policy is also applied to wind power plants already completed and operating in the RM, which when cancelled start operating in the FM. This policy is applied every two years.

The results of the selected policies on the evolution of installed wind capacity compared to the base scenario (Business as Usual – BAU) in the RM are presented in Fig. 5. The results are divided into three categories for better understanding: i) Aggressive Scenarios (C.A.), which represent the best performance; ii) Moderate Scenarios (C.M.), which represent the poor performance; and iii) Soft Scenarios (C.B.), which represent the weakest performance.

Only three of the eight scenarios have a positive effect on the growth of installed wind capacity in the RM compared to the base scenario (i.e.

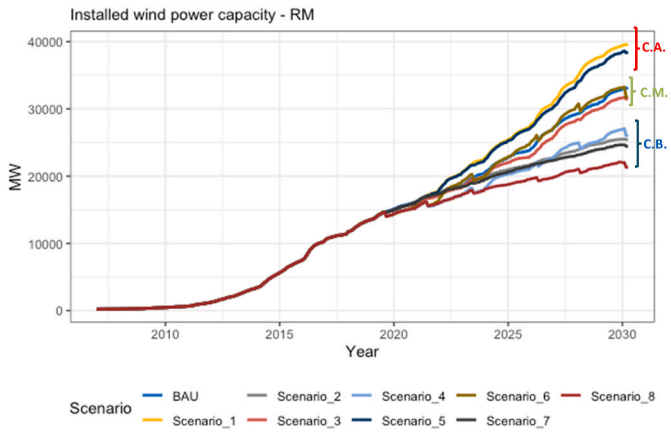

Fig. 5. Evolution of wind capacity installed in the RM in each scenario.

scenarios 1, 5 and 6), which result in the growth of 6492 MW; 5470 MW and 321 MW compared to the base scenario. The other scenarios present inhibition of wind power expansion in values between 1213 MW (scenario 3) and 10,701 MW (scenario 8). At the beginning of 2030, the base scenario shows 32,960 MW and 10,140 MW of installed wind capacity in the RM and FM, respectively. These values are equivalent to 15.5 and 16.0% of the total installed electric capacity in RM and FM, respectively, and 15.6% of the total Brazilian electric capacity.

The effect of policy implementation on the evolution of installed wind capacity in the FTA is presented in Fig. 6. In the FTA, scenario 1 is the only one that does not favour wind deployment. Considering that this is the scenario that most favours wind expansion in RTA, the implementation of policies in this scenario influences the migration of entrepreneurs from FTA to RTA. As for the remaining scenarios, the ones number 6, 8 and 4 produce the best performances, providing growth of 4740 MW, 5040 MW and 3210 MW respectively, about the base scenario.

The difference between the total installed wind capacity in the two contract markets and the base scenario was also analysed, as shown in Fig. 7. In all, three scenarios present an expansion of the total installed wind capacity in the country about the base scenario: scenarios 5, 1 and 6. These scenarios present an increase of 6,270, 5802 and 5252 MW in installed wind capacity at the end of the period analysed. The remaining scenarios present inhibition of wind expansion in values between 450 MW (scenario 3) and 7857 MW (scenario 7). In comparison to the total installed electric generation capacity in Brazil, the results vary between 12.8% (scenario 7) and 17.9% (scenario 5) of wind power generation representativeness at the end of the analysed period.

A summary of the implementation of the policies for each scenario is presented in Fig. 8. The results of the simulations indicate that the

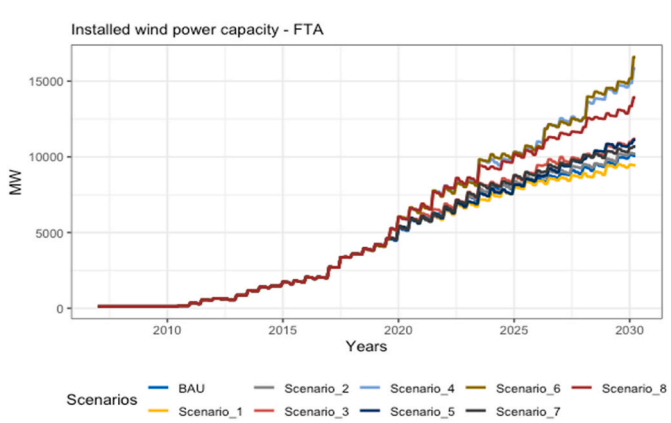

Fig. 6. Evolution of installed wind capacity in the FTA in each scenario.

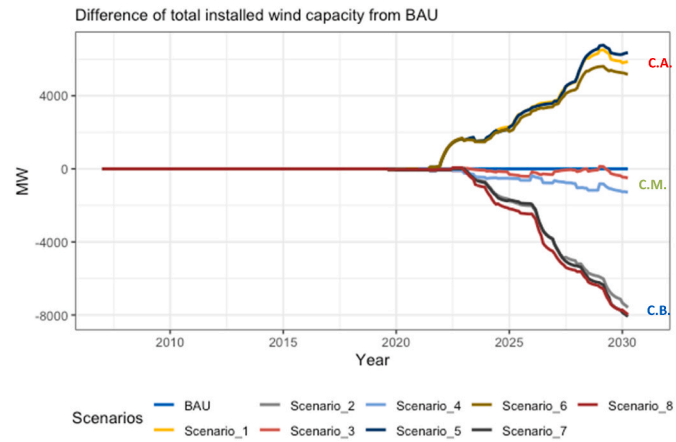

Fig. 7. Difference of total installed wind capacity from base scenario.

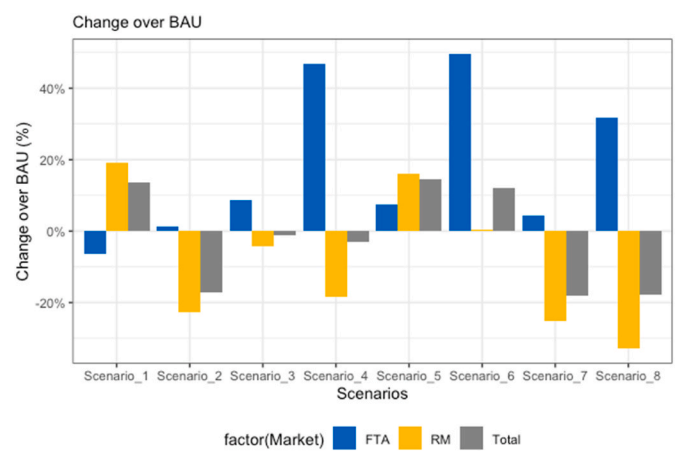

Fig. 8. Summary of results for each scenario in relation to the baseline scenario.

policies involving variations in the contract ceiling prices are the ones that most influence the expansion of installed wind capacity in Brazil. Other incentive policies, which inhibit the development of the wind source in RM, have their potential effects on total wind capacity when combined with project cancellation policies. It should also be noted that the incentive to other sources in the RM causes the migration of some wind projects from this market to the FTA, which becomes a more attractive option in comparison (mainly viewed in scenario 2).

6. Conclusion and policy implications

This paper analysed the effect of guidelines currently applied in Brazilian auctions for the expansion of installed wind capacity in the existing contract markets in Brazil. More specifically, it was explored how the new mechanism of project cancellation and the establishment of different ceiling prices in energy auctions can influence the deployment of wind generation technology in Brazilian procurement markets. The results of the simulations allow the identification, among the policies tested, of incentive and inhibition measures for the deployment of wind capacity installed in RM, FM and both markets simultaneously. The influence of hydrological variations on the establishment of REA guidelines was also verified in the simulations: as drought periods affect water generation, the volume auctioned in REAs tends to be greater. The use of a hybrid simulation model to carry out an analysis of the strategic energy planning sector is still worth mentioning, a sector that still has few reference works using models of this type.

Project cancellation policy has less effect on total installed wind

capacity but impacts more on the development of wind technology in the FTA. Still, concerning this policy, the increase in the cancellation rate of 15% has an incentive effect on the total installed wind capacity less than the use of 2% rates. An interesting result for this purpose can be seen in the application of the policies of scenarios 1, 5 and 6, which involve the use of free competition.

- Although the combination of 2% contract cancellation with free competition in scenario 5 encourages total wind expansion compared to scenario 1, where only free competition is applied, when varied to 15% in scenario 6 the result is lower than in Scenario 1.
- This result indicates that there is a saturation of wind power generation technology in the FTA with the use of a 15% contract cancellation rate together with a free competition policy, which causes an inhibition of the total expansion of this generation technology.
- This saturation occurs when the wind generation expansion in the FTA grows between 46.8% and 49.7% about the base scenario, or the equivalent to a representation of approximately 23% of this type of generation concerning the total generation capacity of the FTA.

Considering the individual application of the policies in Brazil, some findings can still be made:

- The policy applied in scenario 1 is the most effective to encourage wind expansion in RM. On the one hand, the combination of policies applied in scenarios 5 and 6 have a positive impact on the deployment of wind power in Brazil, mainly concentrated in the RCA, scenario 5, and the FTA, scenario 6. (Note that scenarios 5 and 6 are the only ones that present wind expansion in both contract markets.)
- On the other hand, scenarios 3 and 4 based only on the project cancellation mechanism have little effect on the deployment of the wind power, but they have quite an impact on the deployment of this technology in the FTA.
- As mentioned in section 2 of this article, the expansion of wind technology in this market should favour a drop in the price of contract wind technology in the RM due to increased competition, in addition to potentially favouring the rate of innovation of wind generation ventures.

Finally, the policy mix applied in scenarios 2, 7 and 8 inhibits the development of wind power generation at the cost of encouraging other sources. Combinations of this kind are suitable for slowing down wind expansion in the face of possible instabilities generated in the NIS, for example, while contributing to the development of other sources. In this case, the policy mix applied in scenario 8 would have its effects perceived most quickly, since it is the one that most inhibits the growth of the RTA, further positively impacting the development of this source in the FTA.

This paper contributes to a better understanding of the electricity market when there is a project cancellation policy and variation in ceiling prices. One limitation of the study was that the ABM-SD model only used historical data from energy agencies, ignoring the agent's preferences for a specific generation source because of available information. Consequently, future research should consider incorporating more behavioural rules and adaptive behaviour to generate a more comprehensive output.

CRediT authorship contribution statement

Matheus Eduardo Leusin: Conceptualization, Formal analysis, Software, Validation, Writing – original draft. **Mauricio Uriona Maldonado:** Supervision, Conceptualization, Writing – original draft, Writing – review & editing. **Milton M. Herrera:** Supervision, Conceptualization, Data curation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] H. Geller, R. Schaeffer, A. Szklo, M. Tolmasquim, Policies for advancing energy efficiency and renewable energy use in Brazil, Energy Pol. 32 (2004) 1437–1450, https://doi.org/10.1016/S0301-4215(03)00122-8.
[2] M.T. Tolmasquim, T. de Barros Correia, N. Addas Porto, W. Kruger, Electricity market design and renewable energy auctions: the case of Brazil, Energy Pol. 158 (2021), 112558, https://doi.org/10.1016/j.enpol.2021.112558.
[3] M.M. Herrera, F. Cosenz, I. Dyner, How to support energy policy coordination? Findings from the Brazilian wind industry, Electr. J. 32 (2019), https://doi.org/10.1016/j.tej.2019.106636.
[4] B. Bayer, Experience with auctions for wind power in Brazil, Renew. Sustain. Energy Rev. 81 (2018) 2644–2658, https://doi.org/10.1016/j.rser.2017.06.070.
[5] S. Zapata, M. Castaneda, M.M. Herrera, I. Dyner, Investigating the concurrence of transmission grid expansion and the dissemination of renewables, Energy (2023), 127571, https://doi.org/10.1016/j.energy.2023.127571.
[6] S. Zapata, M. Castaneda, C.J. Franco, I. Dyner, Clean and secure power supply: a system dynamics based appraisal, Energy Pol. 131 (2019) 9–21, https://doi.org/10.1016/j.enpol.2019.04.028.
[7] A.A. Juárez, A.M. Araújo, J.S. Rohatgi, O.D.Q. De Oliveira Filho, Development of the wind power in Brazil: political, social and technical issues, Renew. Sustain. Energy Rev. 39 (2014) 828–834, https://doi.org/10.1016/j.rser.2014.07.086.
[8] A. Bradshaw, G. de Martino Jannuzzi, Governing energy transitions and regional economic development: evidence from three Brazilian states, Energy Pol. 126 (2019) 1–11, https://doi.org/10.1016/j.enpol.2018.05.025.
[9] R.C. Da Silva, I. De Marchi Neto, S. Silva Seifert, Electricity supply security and the future role of renewable energy sources in Brazil, Renew. Sustain. Energy Rev. 59 (2016) 328–341, https://doi.org/10.1016/j.rser.2016.01.001.
[10] A.O. Pereira, R. Cunha Da Costa, C.D.V. Costa, J.D.M. Marreco, E.L. La Rovere, Perspectives for the expansion of new renewable energy sources in Brazil, Renew. Sustain. Energy Rev. 23 (2013) 49–59, https://doi.org/10.1016/j.rser.2013.02.020.
[11] J. Schmidt, R. Cancella, A.O.P. Junior, The effect of windpower on long-term variability of combined hydro-wind resources: the case of Brazil, Renew. Sustain. Energy Rev. 55 (2016) 131–141, https://doi.org/10.1016/j.rser.2015.10.159.
[12] J. Schmidt, R. Cancella, A.O. Pereira, An optimal mix of solar PV, wind and hydro power for a low-carbon electricity supply in Brazil, Renew. Energy 85 (2016) 137–147, https://doi.org/10.1016/j.renene.2015.06.010.
[13] G. Aquila, L.C.S. Rocha, P. Rotela Junior, E. de O. Pamplona, A.R. de Queiroz, A. P. de Paiva, Wind power generation: an impact analysis of incentive strategies for cleaner energy provision in Brazil, J. Clean. Prod. 137 (2016) 1100–1108, https://doi.org/10.1016/j.jclepro.2016.07.207.
[14] B. Bayer, L. Berthold, B. Moreno Rodrigo de Freitas, The Brazilian experience with auctions for wind power: an assessment of project delays and potential mitigation measures, Energy Pol. 122 (2018) 97–117, https://doi.org/10.1016/j.enpol.2018.07.004.
[15] Agencia Nacional de Energía Eléctrica-ANEEL, Leilão de geração, 2019.
[16] B.A. Diniz, A. Szklo, M.T. Tolmasquim, R. Schaeffer, Delays in the construction of power plants from electricity auctions in Brazil, Energy Pol. 175 (2023), 113467, https://doi.org/10.1016/j.enpol.2023.113467.
[17] D.B. Carvalho, B.L. Pinto, E.C. Guardia, J.W. Marangon Lima, Economic impact of anticipations or delays in the completion of power generation projects in the Brazilian energy market, Renew. Energy 147 (2020) 1312–1320, https://doi.org/10.1016/j.renene.2019.09.074.
[18] J. Zhao, E. Mazhari, N. Celik, Y.-J. Son, Hybrid agent-based simulation for policy evaluation of solar power generation systems, Simulat. Model. Pract. Theor. 19 (2011) 2189–2205, https://doi.org/10.1016/j.simpat.2011.07.005.
[19] J. Köhler, F. De Haan, G. Holtz, K. Kubeczko, E. Moallemi, G. Papachristos, E. Chappin, Modelling sustainability transitions: an assessment of approaches and challenges, J. OfArtificial Soc. Soc. Simul. 21 (2018), https://doi.org/10.18564/jasss.3629.
[20] S.C. Brailsford, T. Eldabi, M. Kunc, N. Mustafee, A.F. Osorio, Hybrid simulation modelling in operational research: a state-of-the-art review, Eur. J. Oper. Res. 278 (2019) 721–737, https://doi.org/10.1016/j.ejor.2018.10.025.
[21] K. Sharma, J. Al-Zaili, A. Sayma, A System Dynamics Approach to Assess the Impact of Policy Interventions on the Market Penetration of Micro Gas Turbines, 2023, https://doi.org/10.1115/GT2023-101952.
[22] J.D. Sterman, Business Dynamics System Thinking and Modeling for a Complex World, McGraw Hill, Boston, 2000.

[23] M.J. Karimi, S. Vaez-Zadeh, An agent-based model for electric energy policy assessment, Elec. Power Syst. Res. 192 (2021), 106903, https://doi.org/10.1016/j.epsr.2020.106903.
[24] M. Laimon, T. Mai, S. Goh, T. Yusaf, System dynamics modelling to assess the impact of renewable energy systems and energy efficiency on the performance of the energy sector, Renew. Energy 193 (2022) 1041–1048, https://doi.org/10.1016/j.renene.2022.05.041.
[25] A.A. Romano, G. Scandurra, A. Carfora, M. Fodor, Renewable investments: the impact of green policies in developing and developed countries, Renew. Sustain. Energy Rev. 68 (2017) 738–747, https://doi.org/10.1016/j.rser.2016.10.024.
[26] R.M. Dutra, A.S. Szklo, Incentive policies for promoting wind power production in Brazil : scenarios for the Alternative Energy Sources Incentive Program (PROINFA) under the New Brazilian electric power sector regulation, Renew. Energy 33 (2008) 65–76, https://doi.org/10.1016/j.renene.2007.01.013.
[27] Z. Abdmouleh, R.A.M. Alammari, A. Gastli, Review of policies encouraging renewable energy integration & best practices, Renew. Sustain. Energy Rev. 45 (2015) 249–262, https://doi.org/10.1016/j.rser.2015.01.035.
[28] P. del Río, P. Linares, Back to the future? Rethinking auctions for renewable electricity support, Renew. Sustain. Energy Rev. 35 (2014) 42–56, https://doi.org/10.1016/j.rser.2014.03.039.
[29] J. Kreiss, K.-M. Ehrhart, M.-C. Haufe, Appropriate design of auctions for renewable energy support – prequalifications and penalties, Energy Pol. 101 (2017) 512–520, https://doi.org/10.1016/j.enpol.2016.11.007.
[30] P.R. Gribik, Designing an auction for QF generation resources in California: what went wrong? Electr. J. 8 (1995) 14–23, https://doi.org/10.1016/1040-6190(95)90197-3.
[31] P. Mastropietro, C. Batlle, L.A. Barroso, P. Rodilla, Electricity auctions in South America: towards convergence of system adequacy and RES-E support, Renew. Sustain. Energy Rev. 40 (2014) 375–385, https://doi.org/10.1016/j.rser.2014.07.074.
[32] R.P. Mcafee, J. Mcmillan, M. Carter, K. Harley, P. Howitt, P. Kuhn, D. Laidler, G. Macdonald, R. Porter, W. Samuelson, J. Whalley, Auctions and bidding, J. Econ. Lit. 25 (1987) 699–738.
[33] Y. Atalay, A. Kalfagianni, P. Pattberg, Renewable energy support mechanisms in the Gulf Cooperation Council states: analyzing the feasibility of feed-in tariffs and auction mechanisms, Renew. Sustain. Energy Rev. 72 (2017) 723–733, https://doi.org/10.1016/j.rser.2017.01.103.
[34] A. Bradshaw, Regulatory change and innovation in Latin America: the case of renewable energy in Brazil, Util. Pol. 49 (2017) 156–164, https://doi.org/10.1016/j.jup.2017.01.006.
[35] R. Moreno, L.A. Barroso, H. Rudnick, S. Mocarquer, B. Bezerra, Auction approaches of long-term contracts to ensure generation investment in electricity markets: lessons from the Brazilian and Chilean experiences, Energy Pol. 38 (2010) 5758–5769, https://doi.org/10.1016/j.enpol.2010.05.026.
[36] Observatório do Setor Elétrico Brasileiro, Post-auctions analysis: 19º e 20º Leilão de Energia Existente, 2018.
[37] M.M. Herrera, I. Dyner, F. Cosenz, Assessing the effect of transmission constraints on wind power expansion in northeast Brazil, Util. Pol. 59 (2019), 100924, https://doi.org/10.1016/j.jup.2019.05.010.
[38] M.M. Herrera, M. Uriona, I. Dyner, Dynamics performance of the wind-power supply chain with transmission capacity constraints, Int. J. Electr. Comput. Eng. 10 (2020) 1142–1148, https://doi.org/10.11591/ijece.v10i1.pp1142-1148.
[39] G. Aquila, E. de O. Pamplona, A.R. de Queiroz, P. Rotela Junior, M.N. Fonseca, An overview of incentive policies for the expansion of renewable energy generation in electricity power systems and the Brazilian experience, Renew. Sustain. Energy Rev. 70 (2017) 1090–1098, https://doi.org/10.1016/j.rser.2016.12.013.
[40] E.E. Rego, C. de Oliveira Ribeiro, Successful Brazilian experience for promoting wind energy generation, Electr. J. 31 (2018) 13–17, https://doi.org/10.1016/j.tej.2018.02.003.
[41] C.J. Franco, M. Castaneda, I. Dyner, Simulating the new British electricity-market reform, Eur. J. Oper. Res. 245 (2015) 273–285, https://doi.org/10.1016/j.ejor.2015.02.040.
[42] B.K. Bala, F.M. Arshad, K.M. Noh, System Dynamics: Modelling and Simulation, 2017, https://doi.org/10.1007/SpringerReference_7284.
[43] D. Wheat, What can system dynamics learn from the public policy implementation literature? Syst. Res. Behav. Sci. 27 (2010) 425–442, https://doi.org/10.1002/sres.1039.
[44] J.P. Davis, C.B. Bingham, Developing theory through simulation methods, Acad. Manag. Rev. 32 (2007) 480–499.
[45] C. Swinerd, K.R. McNaught, Design classes for hybrid simulations involving agent-based and system dynamics models, Simulat. Model. Pract. Theor. 25 (2012) 118–133, https://doi.org/10.1016/j.simpat.2011.09.002.
[46] G. Wallentin, C. Neuwirth, Dynamic hybrid modelling: switching between AB and SD designs of a predator-prey model, Ecol. Model. 345 (2017) 165–175, https://doi.org/10.1016/j.ecolmodel.2016.11.007.
[47] G. Schweiger, H. Nilsson, J. Schoeggl, W. Birk, A. Posch, Modeling and simulation of large-scale systems: a systematic comparison of modeling paradigms, Appl. Math. Comput. 365 (2020), 124713, https://doi.org/10.1016/j.amc.2019.124713.
[48] A.P.G. Scheidegger, T.F. Pereira, M.L.M. de Oliveira, A. Banerjee, J.A. B. Montevechi, An introductory guide for hybrid simulation modelers on the primary simulation methods in industrial engineering identified through a systematic review of the literature, Comput. Ind. Eng. 124 (2018) 474–492, https://doi.org/10.1016/j.cie.2018.07.046.
[49] E. Nabavi, K.A. Daniell, H. Najafi, Boundary matters: the potential of system dynamics to support sustainability? J. Clean. Prod. (2016) https://doi.org/10.1016/j.jclepro.2016.03.032.
[50] P. Hansen, X. Liu, G.M. Morrison, Agent-based modelling and socio-technical energy transitions: a systematic literature review, Energy Res. Social Sci. 49 (2019) 41–52, https://doi.org/10.1016/j.erss.2018.10.021.
[51] H. Rahmandad, N. Repenning, J. Sterman, Effects of feedback delays on learning, Syst. Dynam. Rev. 25 (2009) 309–338, https://doi.org/10.1002/sdr.427.
[52] K. Thompson, P. Reimann, Patterns of use of an agent-based model and a system dynamics model: the application of patterns of use and the impacts on learning outcomes, Comput. Educ. 54 (2010) 392–403, https://doi.org/10.1016/j.compedu.2009.08.020.
[53] T. Eldabi, S. Brailsford, A. Djanatliev, M. Kunc, N. Mustafee, A.F. Osorio, Hybrid simulation challenges and opportunities: a life-cycle approach, in: 2018 Winter Simul. Conf., 2018, pp. 1500–1514, https://doi.org/10.1109/WSC.2018.8632465.
[54] A. Alfaris, A. Khiyami, A. Alawad, A. Alsaati, M. Hadhrawi, The integrated energy decision support system, Syst. Eng. 18 (2015) 511–529, https://doi.org/10.1002/sys.21326.
[55] Y. Li, P. Han, J. Wang, X. Song, Modeling and optimization oriented to the micro-grid-EV joint system, in: 2016 12th Int. Conf. Nat. Comput. Fuzzy Syst. Knowl. Discov., 2016, pp. 336–341.
[56] E. Mazhari, J. Zhao, N. Celik, S. Lee, Y.-J. Son, L. Head, Hybrid simulation and optimization-based design and operation of integrated photovoltaic generation, storage units, and grid, Simulat. Model. Pract. Theor. 19 (2011) 463–481, https://doi.org/10.1016/j.simpat.2010.08.005.
[57] F.M. Bass, A new product growth for model consumer durables, Manag. Sci. 15 (1969) 215–227.
[58] L.-B. Desroches, K. Garbesi, C. Kantner, R. Van Buskirk, H.-C. Yang, Incorporating experience curves in appliance standards analysis, Energy Pol. 52 (2013) 402–416, https://doi.org/10.1016/j.enpol.2012.09.066.
[59] M. Jimenez, C.J. Franco, I. Dyner, Diffusion of renewable energy technologies: the need for policy in Colombia, Energy 111 (2016) 818–829, https://doi.org/10.1016/j.energy.2016.06.051.
[60] P. De Jong, A. Kiperstok, A.S. Sánchez, R. Dargaville, E.A. Torres, Integrating large scale wind power into the electricity grid in the Northeast of Brazil, Energy 100 (2016) 401–415, https://doi.org/10.1016/j.energy.2015.12.026.
[61] R. Miranda, R. Soria, R. Schaeffer, A. Szklo, L. Saporta, Contributions to the analysis of "Integrating large scale wind power into the electricity grid in the Northeast of Brazil" [Energy 100 (2016) 401–415], Energy 118 (2017) 1198–1209, https://doi.org/10.1016/j.energy.2016.10.138.
[62] M.M. Herrera, I. Dyner, F. Cosenz, Benefits from energy policy synchronisation of Brazil's North-Northeast interconnection, Renew. Energy 162 (2020) 427–437, https://doi.org/10.1016/j.renene.2020.08.056.
[63] ABEEólica, Annual Wind Energy Report 2017, 2018, p. 27.